Application: From Flavour Chemistry to the Bake

The hybrid dough, pizza, bread and oven spring:
Part III of the notes

Sourdough Science

github.com/zrrbite/sourdough-science

What this deck covers

Practical questions, answered from Part III of the notes:

The hybrid dough

Why leaven with commercial yeast and add a little levain? Where do the fruity esters come from, and what does the oven do to them?

Pizza

Why is a longer ferment not always better? Why does pizza burn, and how do you run a 12” Ooni?

Bread

How do you get the benefits of proteolysis without a long ferment? Which lever matters most?

Handling and oven spring

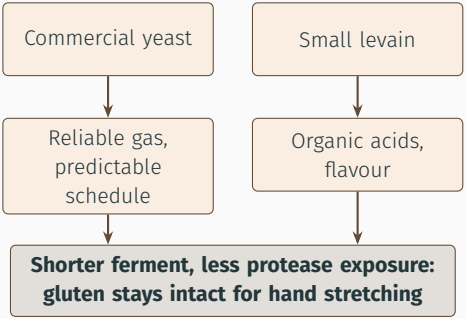
What do you do with dough straight from the fridge? What, besides gas, decides how much it rises?

Source: chapters 12–20 of the notes, and the note *Accelerating Proteolysis: pH, Heat, and Salt*

The hybrid and its three flavour systems

Let commercial yeast leaven and the levain flavour

- In 100% sourdough the wild yeast does all the leavening
- That needs a long ferment, so LAB proteases work on the gluten for longer
- Result: fragile, tear-prone dough that is hard to time
- Hybrid: commercial *S. cerevisiae* makes the gas; a small levain brings acid and flavour



In the kitchen: treat leavening and flavour as two separate jobs and tune each on its own. The levain works even at a small percentage.

Source: ch. 12

Esters are made by enzymes, not by acid meeting alcohol

- Acid + alcohol → ester + water is the minor route in dough: very slow without a catalyst, and the excess water pushes it back (general chemistry; no measurement in dough found)
- Yeast enzymes make their acyl donor inside the cell: the levain's acids are not the feedstock, the alcohols are

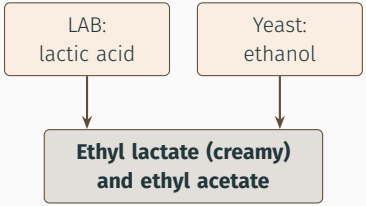
Yeast genes	Join	To give
ATF1, *ATF2*	acetyl-CoA + an alcohol	ethyl acetate (fruity, solvent-like), isoamyl acetate (banana), 2-phenylethyl acetate (rose, honey)
EHT1, *EEB1*	fatty acyl-CoA + ethanol	ethyl hexanoate (apple), ethyl octanoate

In the kitchen: any dough with active yeast makes these esters, levain or not. More fusel alcohol tends to mean more acetate esters; the genes vary by strain.

Source: Verstrepen et al. (2003); Saerens et al. (2010); Pradal et al. (2025); ch. 13

Some sourdough LAB make esters too, with the yeast's help

- In a medium built like wheat sourdough, strains of several LAB species made ethyl acetate and ethyl lactate
- Output was limited by the supply of precursors
- Esterase genes vary by species; the *Fructilactobacillus sanfranciscensis* strains tested made neither
- Ethyl lactate is lactic acid + ethanol, so this route plausibly needs both partners



LAB enzyme, during fermentation: the one route close to "acid + alcohol"

In the kitchen: the levain may add ethyl lactate and more ethyl acetate, but not every levain carries LAB that make them.

Source: Pradal et al. (2025); ch. 13

The oven keeps or loses esters; it does not make them

- Esters and ethanol are volatile: the crumb stays near 98°C while water leaves through a crust past 150°C, carrying them with it
- Crust aroma is Maillard and Strecker chemistry: pyrazines, furanones, Strecker aldehydes, melanoidins
- Pumpernickel: aroma formed mostly in baking (malty 3-methylbutanal, 2-methylbutanal, potato-like methional); ester levels were set by the fermentation (hours at low heat: extreme, but the same way)
- No source was found for esters forming in the crust

In the kitchen: the fruity character is set while the yeast works. A shorter ferment may keep more of it, since CO_2 leaving the dough during folds and degassing plausibly carries esters away.

hypothesis

Source: Majcher et al. (2020); ch. 13

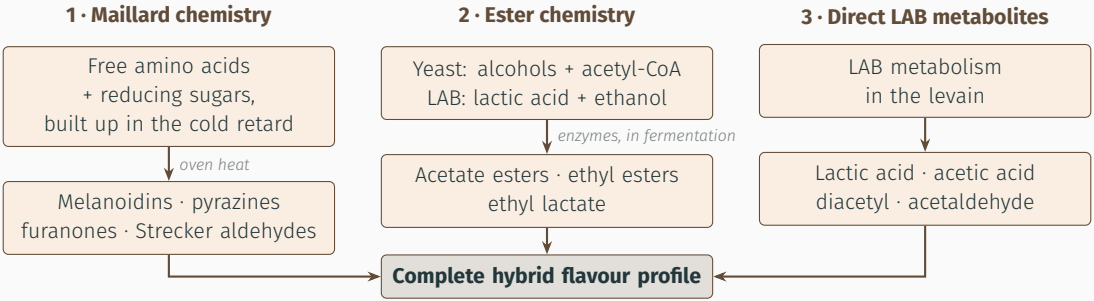
A long cold ferment's complexity is Maillard, not esters

- 48–72 h at 3–5° C: flour enzymes keep working, at reduced but significant rates
- Amylases turn damaged starch into maltose and glucose: roughly 10–15% of room-temperature activity, but 48 h adds up
- Flour proteases cut gluten and other proteins into free amino acids and short peptides
- In the oven these react: melanoidins (crust colour), pyrazines (roasted, nutty), furanones (caramel-like), Strecker aldehydes (malty, bready, chocolatey)
- Yeast's own acids are millimolar, LAB's tens of millimolar: far less sour. Its esters give only a low baseline

In the kitchen: the Maillard gain comes from time and enzyme breakdown, so a long retard pays off whichever organism does the leavening.

Source: ch. 14

The hybrid stacks three flavour systems in one dough

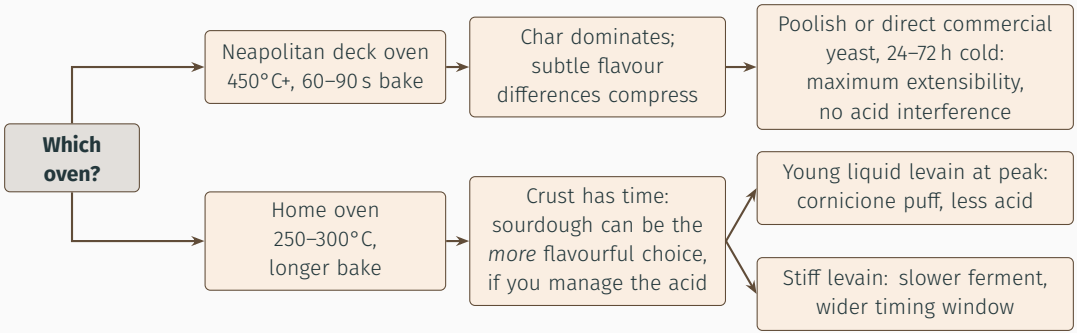


In the kitchen: commercial yeast alone gives system 1 and the yeast half of 2. Pure sourdough gives all three, but with a smaller yeast population, a longer ferment that may lose more volatiles, and structural and timing penalties. The hybrid keeps all three, with structure and schedule intact.

Source: ch. 14

Pizza: the dough, the long ferment and the Ooni

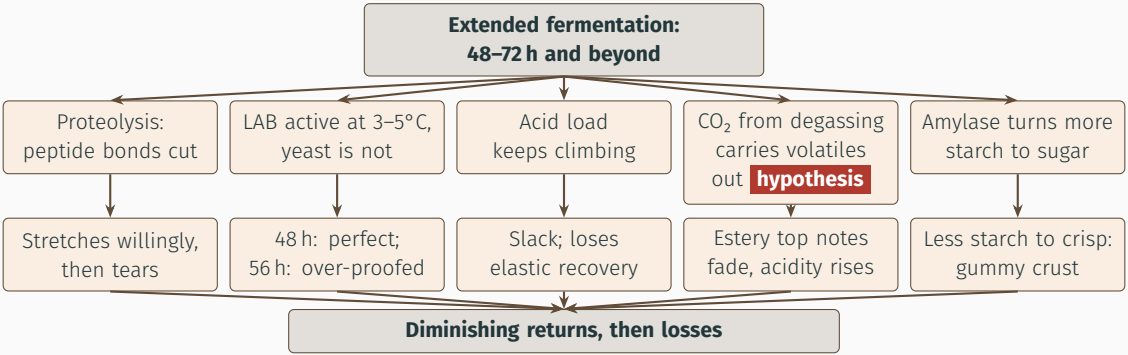
The oven decides which preferment suits a pizza



In the kitchen: acid-modified gluten is more extensible but less elastic: less snap-back, less resilience if overworked. Reduce hydration slightly or raise flour protein.

Source: ch. 15

Longer is not automatically better for pizza



In the kitchen: many excellent pizzas use 48–72 h cold doughs, but longer is not better by default. A shorter, controlled ferment with a hybrid's flavour can be more reliable, and arguably more complex.

Source: ch. 16; CO_2 stripping of volatiles is a hypothesis (ch. 13)

Over-fermented pizza dough fails suddenly, not gradually

- LAB proteases, and to a lesser extent flour proteases, keep cutting the gluten network
- A loaf can use that; a pizza must survive being stretched by hand into a thin, continuous sheet
- Degraded dough stretches willingly until it suddenly doesn't: thin spots and holes, worst in the thin centre
- Acid: moderate acidity helps stretching, too much turns it slack; acetic acid feels tight, then suddenly gives way
- LAB keep working at 3–5 °C while yeast nearly stops: perfect at 48 h, over-proofed at 56 h

In the kitchen: with sourdough the usable window is narrow and hard to predict. Low-dose commercial yeast in a 48–72 h cold ferment can give a window several hours wide.

Source: ch. 16

Residual sugar, not hydration, is why pizza burns

Ferment	Sugar at launch	At 450°C+
Short	yeast has not eaten the sugar pool	uniform carbonisation in seconds
24–72 h cold retard	yeast has eaten most simple sugars	controlled leopard spots
Too long	amylase outruns the yeast: sugars re-accumulate	burns *and* handles poorly

- Flour has about 1.5–2% free sugars, plus damaged starch that amylases turn into maltose and glucose
- Water pins the surface near 100°C until it dries: a large buffer at 250–300°C, mere seconds at 450°C+

In the kitchen: fix burning through the ferment, not the water: more hydration buys seconds but cannot rescue a sugar-rich dough.

Source: ch. 18

In a 12” Ooni, two ball sizes both work

	Larger ball: forgiving	Smaller ball: crisp
Ball → pie	270–280 g → 10–11”	220–230 g → 11–12”
Crust	thicker, more thermal mass	thinner, cooks through fast
At 40–50% flame	2–3 min bake	90 s to 2 min bake
Margin for error	wide	small: done to carbonised in seconds
Maillard aroma	more time	less time
Eats like	bread-tender, crisp outside; Roman/NY	cracker-crisp, no raw centre

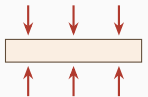
- Hydration 63–67%: above about 68–70% the dough sticks on launch and sags, and if it folds on the stone there is no room to recover

In the kitchen: the choice is stylistic. Both want a 48 h cold retard (sugar depletion) and a moderate 40–50% of max flame.

Source: ch. 18

Twice the crust thickness, four times the time to cook through

- Heat comes from stone and flame and diffuses slowly inward
- Neapolitan crust (~3 mm): gelatinised (60–80°C) right through, well within 60 s
- 275 g ball at 10–11”: ~6–8 mm, the path roughly doubles
- Time scales with distance squared (Fick’s law): about 4×
- Full flame chars the surface first: burnt outside, gummy or raw inside



~3 mm
through in seconds

~6–8 mm
2× the path, ~4× the time

flame above, stone below

In the kitchen: trade surface heat for diffusion time: lower flame, longer bake, so the centre cooks before the outside chars.

Source: Sadot et al. (2017); ch. 18

Ooni 12”: preheat hard, bake moderate, boost at the end

Step	Flame	Target
Preheat	full, about 15 min	stone heat-soaked; IR reading 380–420°C. Longer risks an overheated thin stone
Pre-launch	40–50% of max	stone 350–380°C; check 30–60 s after the flame settles
Bake	40–50%	2–3 min, turning every 20–30 s
Char boost	briefly up	final 15–20 s, optional: leopard spots on the cornicione
Recovery	full	stone drops 50–100°C per pizza; 3–5 min back to target, then 40–50%

In the kitchen: this suits the hybrid. The retard leaves enough sugar to brown but not burn; 2–3 min gives more Maillard and Strecker time than a 60 s bake; the esters were set in fermentation. A cold stone gives a pale, soggy bottom.

Source: ch. 18

Bread: controlled proteolysis

For bread, proteolysis is a tool rather than a failure

	Pizza	Bread
Failure mode	tears when stretched to a 3–6 mm sheet	loses gas retention: flat, dense loaf
Threshold	near	much further out
Tolerable levain	10–15%	20–30%

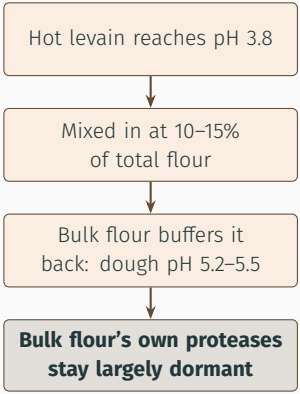
- **Extensibility:** easier shaping, more oven spring before the network resists
- **Softer crumb:** partly depolymerised glutenin bites less chewy
- **Free amino acids:** Maillard substrate, fully expressed in a 40–50 min bake, near-irrelevant in a 2–3 min pizza bake

In the kitchen: a loaf takes far more proteolysis than a pizza before it fails: use it on purpose.

Source: Bombara et al. (1997); ch. 17; note *Accelerating Proteolysis*

Acidifying the levain barely acidifies the dough

- Cereal aspartic proteases peak around pH 3.5–4.0
- Buffering is molar: pH 6.0 → 4.0 needs a fixed quantity of acid, however fast it is made
- Hot LAB (40–45°C) acidify perhaps 2–4× faster: 8 h might become 2–3 h, not 30 min
- Higher-ash flour buffers harder: its minerals resist the drop that would activate its proteases



In the kitchen: getting the levain acidic is easy; getting the final dough to pH 3.5–4.0 quickly is not, and heat alone will not do it.

Source: Thiele et al. (2004); ch. 17

In a hot bulk, heat does the work that pH cannot

- Enzyme rates roughly double for every 10°C rise ($Q_{10} \approx 2$), up to denaturation
- Wheat proteases stay stable to about 50°C
- Further from the pH optimum but nearer the thermal optimum: the two effects partly cancel
- Not a dramatic acidification: ordinary proteases running hot

pH 4.5
25°C
≈
pH 5.5
40°C

comparable total proteolytic throughput

Caveat: $Q_{10} \approx 2$ is a general biochemical approximation, not a bread measurement. Use it as an order-of-magnitude guide.

In the kitchen: don't chase a low pH; use the temperature axis of the enzyme curve instead.

Source: Bloksma & Bushuk (1988); ch. 17

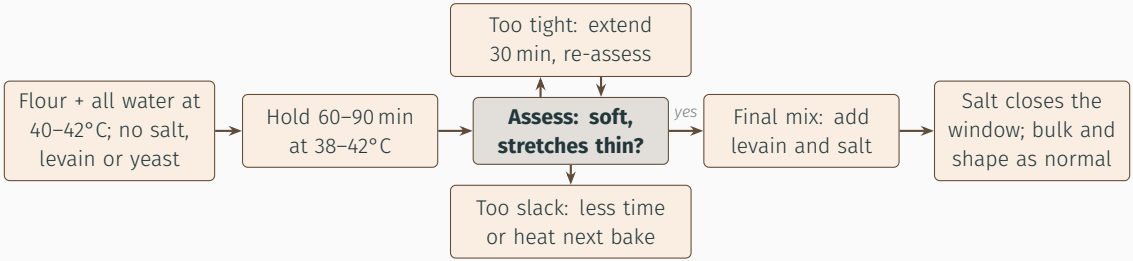
Salt is the biggest brake on proteolysis

- Salt suppresses cereal proteases two ways: higher ionic strength (enzyme meets substrate less well) and lower water activity (less hydrolysis)
- It works regardless of pH, and the effect is substantial at normal bread levels of 1.8–2.2% of flour
- An unsalted dough lets its proteases run considerably faster
- Calvel's autolyse is usually explained as hydration. That is correct but incomplete: part of its softening is enzymatic

In the kitchen: hold the salt back until the final mix. It is a bigger proteolytic lever than any realistic pH change, costs nothing, and is reversed by salting.

Source: Beck et al. (2012); Calvel et al. (2001); ch. 17

A warm autolyse gives proteolysis in an hour, no fermentation



- Proteases at pH ~6.0 but 2–4× faster (Q_{10}), no salt: no acid, buffering or microbes to manage
- Stay below about 50°C: starch gelatinises from 55–60°C, making a scald (*Brühstück*) that denatures the proteases
- Thermal lag: a large mass in a plastic tub takes 30–45 min to reach the setpoint at its core

Source: Eliasson (1980); ch. 17. **not yet tested** against a room-temperature control

Rank the levers: salt first, pH last

	Lever	Why
1	Remove salt from the autolyse	the largest single effect; free; reversed by salting at the final mix
2	Raise autolyse or bulk to 38–42°C	doubles to quadruples enzyme rate via Q_{10}, no pH change needed
3	Increase the levain percentage	moves acid and LAB protease load together; bread tolerates 20–30%, but the loaf gets more acidic
4	Extend time	always works, but it is what we set out to avoid
5	Manipulate pH directly	least efficient via fermentation: buffering and dilution

In the kitchen: with fermentation and proteolysis decoupled, the ferment can be as long or as short as the flavour goal needs.

Source: ch. 17

Direct acidification: a shortcut that tightens before it softens

- A little food-grade lactic acid reaches the target pH at once, with no wait for LAB
- With a 40°C hold: rapid proteolysis with no extended fermentation
- But added acid first firms the dough and cuts extensibility (charge-state and hydration changes)
- Held warm, enzymatic softening catches up and overtakes; expect a phase where the dough seems to go the wrong way
- Chemical and biological acidification do not give the same rheology: the literature here is contested **contested**

In the kitchen: a specialist tool. For most bread, the no-salt warm autolyse gives enough proteolysis with fewer variables and no risk of a dough that cannot hold gas.

Source: Clarke et al. (2004); Wehrle et al. (1997); ch. 17; ch. 25

The promised flavour dividend is the least-evidenced claim

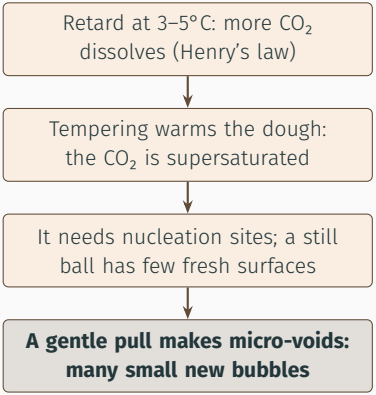
- The claim: the warm autolyse enlarges the free amino acid pool
- In a 40–50 min bake that pool feeds Strecker degradation
- Expected result: malty, nutty and roasted crust notes from Strecker aldehydes and pyrazines
- The payoff a short pizza bake cannot access

not yet tested The repo calls this the least directly evidenced claim in the document. It wants a sensory comparison: warm-autolyse loaf against a control on the same formula.

Source: Strecker chemistry: Yang et al. (2022); claim: ch. 17 and note *Accelerating Proteolysis*; status: README

Handling and oven spring

After the retard, pull the dough; don't press it



- Like tapping a carbonated drink: disturbance triggers nucleation, not more yeast activity
- The pull also re-tensions the gluten, which holds the bubbles and lifts vertically
- Pressing flat expels gas; sluggish post-retard yeast must re-inflate from scratch
- Pull *adds* gas; press *removes* it

In the kitchen: lift an edge, stretch without tearing, fold over the centre; rotate 90° and repeat 2–3 times. Re-round loosely, don't degas. Expect fast rise in 20–30 min; tempering stays 1.5–2 h.

Source: ch. 19; CO_2 partitioning: Chiotellis & Campbell (2003)

Same dough, three handlings, three different rises own result

	Retarded in	Then	Result
1	wide, flat container	turned out, cut into rolls	good rise
2	bowl	turned out, cut into rolls	less rise
3	wide (dough of 1)	flattened like a pizza centre, slapped, baked flat	great rise, open crumb

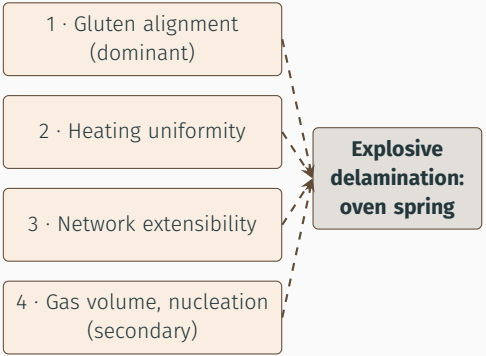
- Same formula, retard and yeast activity: only the mechanical state of the gluten changed
- Wide container: dough spreads under gravity, strands align horizontally (laminar), gas finds an easier path upward
- Bowl: a tangled, random mesh with a compressed bottom, resisting expansion in every direction

In the kitchen: retard in a wide, flat container rather than a bowl.

Source: ch. 20; the author's own experiments

Steam delamination: gluten alignment beats gas volume

- A thin disc reaches steam temperature right through at once; a roll heats as a slow wave
- Water turns to steam, about 1,600-fold in volume
- Steam follows the weak interfaces between aligned gluten planes and pushes them apart in one event: pita physics
- The holes are gaps between separated layers, not coalesced fermentation bubbles



In the kitchen: a well-aligned, largely degassed dough outperforms a gassy dough with random gluten orientation. Thinner pieces spring more per gram.

Source: ch. 20; the hierarchy is drawn from the author's own experiments **own result**

Fold to multiply the layers, then rest 15–30 minutes

Fold

- Letter fold in thirds, rotate 90°, flatten, fold again: 2 cycles ~9 layers, 3 cycles ~27
- Puff pastry without the fat: rougher open crumb, not flakes
- Plain dough blurs at about 20–40 effective layers (puff pastry: 700–900)

Rest

- Just folded, the gluten is elastic and would contract against the steam
- Resting relaxes it (thiol–disulfide interchange) while CO_2 nucleates into the new interfaces
- Beyond 30–40 min the layers merge and the dough spreads

In the kitchen: for rolls or flatbreads: turn out from a wide container, flatten, letter-fold twice, portion loosely, rest 15–20 min until slightly puffed, bake at high heat.

Source: ch. 20

In pizza, shaping makes the cornicione puff

- Hand-stretching aligns the gluten laminarly in the rim; oven steam then delaminates those layers
- A well-aligned rim on slightly under-proofed dough out-puffs a poorly handled rim on perfectly proofed dough
- Neapolitan slapping is not just shaping the disc: it builds that alignment
- The pressed-thin centre has too little cross-section to puff; the rim has both alignment and thermal mass

In the kitchen: for the hybrid dough in an Ooni: gentle pull, temper 1.5–2 h, then open with deliberate attention to the rim's alignment. Shaping matters more than fermentation state for the puff.

Source: ch. 20, applying the author's roll and flatbread experiments to pizza

Takeaways

- Let commercial yeast leaven and a small levain flavour: three flavour systems, intact gluten, a predictable schedule
- Esters are made by enzymes while the yeast works; the oven only keeps or loses them. Crust aroma is Maillard and Strecker chemistry
- For pizza, longer is not automatically better: proteolysis, acid, a narrowing window and returning sugar all turn against you
- Burning is residual sugar. In a 12” Ooni: 48 h retard, 63–67% hydration, 40–50% flame, 2–3 min on a 350–380°C stone
- For bread proteolysis, hold back the salt and warm the autolyse to 38–42°C (below 50°C) before chasing pH
- After the retard, pull, don’t press. For oven spring, gluten alignment beats gas volume

Source: chapters 12–20

Open questions

- Warm autolyse against a room-temperature control on the same formula: extensibility by hand, crumb softness after the bake **not yet tested**
- Crust aroma, warm-autolyse loaf vs control: a sensory check of the Strecker/pyrazine claim, the least directly evidenced in the document **not yet tested**
- Log the core dough temperature during the warm autolyse to measure the thermal lag **not yet tested**
- Direct lactic acid + 40°C hold as a comparison arm: expect tightening before the enzymes soften it **not yet tested**
- Does CO_2 leaving during folds and degassing strip esters? No measurement in dough was found **hypothesis**

Source: note *Accelerating Proteolysis* (open threads); README (status); ch. 13; ch. 16

Sources

Esters and fermentation volatiles

- **Verstrepen KJ et al. (2003)** Expression levels of the yeast alcohol acetyltransferase genes *ATF1*, *Lg-ATF1*, and *ATF2* control the formation of a broad range of volatile esters. *Appl Environ Microbiol* 69(9):5228–5237.
- **Saerens SMG, Delvaux FR, Verstrepen KJ, Thevelein JM (2010)** Production and biological function of volatile esters in *Saccharomyces cerevisiae*. *Microb Biotechnol* 3(2):165–177.
- **Pradal I, Weckx S, De Vuyst L (2025)** The production of esters by specific sourdough lactic acid bacteria species is limited by the precursor concentrations. *Appl Environ Microbiol* 91(3):e0221624.
- **Majcher MA et al. (2020)** Formation of key aroma compounds during preparation of pumpernickel bread. *J Agric Food Chem* 68(38):10352–10360.
- **Yang Y et al. (2022)** Research progress on the formation mechanism and detection technology of bread flavor. *J Food Sci* 87(8):3330–3351.

Gas and heat

Sources (cont.)

- **Chiotellis E, Campbell GM (2003)** Proving of bread dough I: modelling the evolution of the bubble size distribution. *Food Bioprod Process* 81(3):194–206.
- **Sadot M et al. (2017)** Numerical modelling of convective heat transfer during bread baking. *J Food Eng*.

Acidification, proteolysis, salt and autolyse

- **Thiele C, Grassl S, Gänzle MG (2004)** Gluten hydrolysis and depolymerization during sourdough fermentation. *J Agric Food Chem* 52(5):1307–1314.
- **Clarke CI et al. (2004)** Wheat sourdough fermentation: effects of time and acidification on fundamental rheological properties. *Cereal Chem* 81(3):409–417.
- **Wehrle K, Grau H, Arendt EK (1997)** Effects of lactic acid, acetic acid, and table salt on fundamental rheological properties of wheat dough. *Cereal Chem* 74(6):739–744.
- **Calvel R, Wirtz RL, MacGuire JJ (2001)** *The Taste of Bread*. Aspen Publishers.
- **Beck M, Jekle M, Becker T (2012)** Impact of sodium chloride on wheat flour dough for yeast-leavened products. *J Sci Food Agric* 92(2):299–306.

Sources (cont.)

- **Bombara N, Añón MC, Pilosof AMR (1997)** Functional properties of protease modified wheat flours. *LWT* 30(5):441–447.
- **Bloksma AH, Bushuk W (1988)** Rheology and chemistry of dough. In *Wheat: Chemistry and Technology*, 3rd ed., AACC.
- **Eliasson AC (1980)** Effect of water content on the gelatinization of wheat starch. *Starch/Stärke* 32(8):270–272.

Full list: ch. 25 of the notes.